

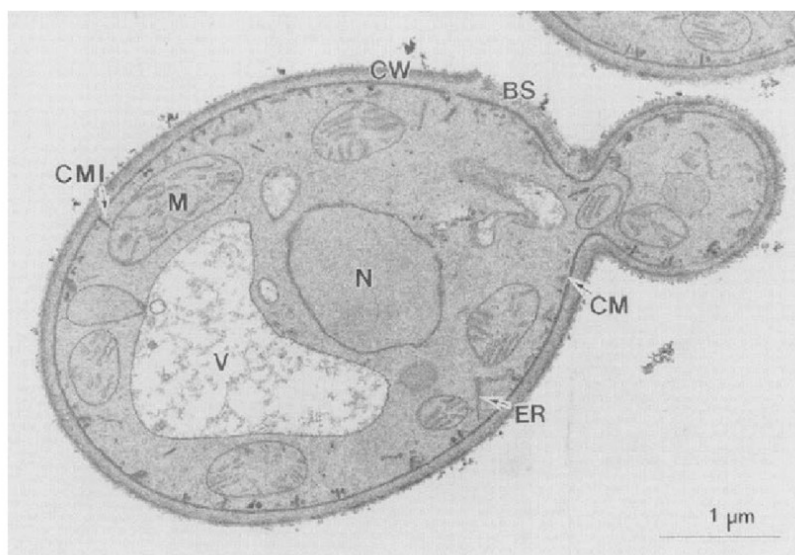
Section I

20 marks

Attempt Questions 1 – 20.

Complete the multiple choice answer sheet on page 11.

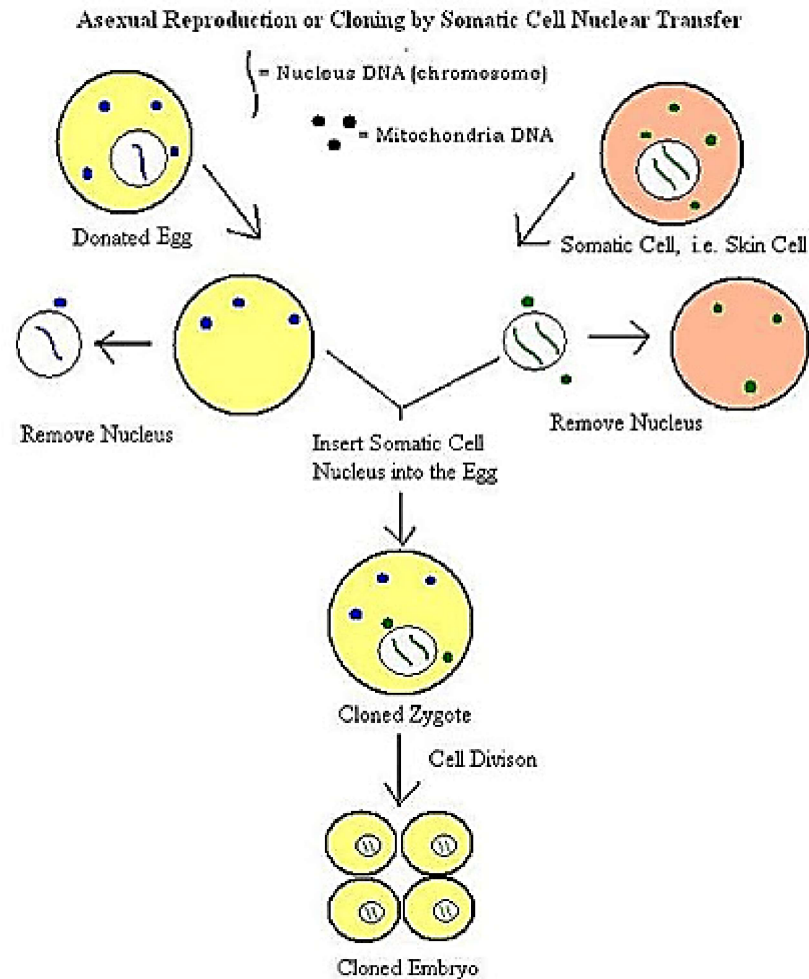
- 1 The nitrogenous base uracil pairs with which other base?
- A. Adenine
 - B. Cytosine
 - C. Guanine
 - D. Thymine
- 2 Examine the following diagram of a pathogen undergoing a reproductive process:



What type of pathogen is it most likely to be?

- A. Virus
- B. Bacteria
- C. Fungus
- D. Protist

- 3 Koch's postulates can be used to determine:
- A. The most appropriate vaccination to combat a pathogen
 - B. How effective a public health campaign is at reducing the risk of contracting a disease
 - C. Whether a pathogen is responsible for causing a particular disease
 - D. The mode of transmission for a pathogen
- 4 Artificial pollination involves transferring pollen from one flower:
- A. to the anther of another flower.
 - B. to the ovary of another flower
 - C. to the stamen of another flower.
 - D. to the stigma of another flower
- 5 An infectious disease was found to be effectively controlled using improved water treatment methods.
What is the most likely method of transmission for this disease?
- A. Direct
 - B. Indirect
 - C. Vector
 - D. Hereditary
- 6 Red-green colour blindness is an X-linked trait. A woman who is red-green colour blind has a son with a man who has normal colour vision.
What is the chance that their son will have red-green colour blindness?
- A. 0%
 - B. 25%
 - C. 50%
 - D. 100%
- 7 In 1996, somatic cell nuclear transfer produced the first artificially cloned mammal, Dolly the sheep.



Which statement about Dolly's complete genome is true?

- A. Dolly was completely genetically identical to the sheep that provided the donor egg.
- B. Dolly was completely genetically identical to the sheep that provided the donated nucleus.
- C. Dolly was completely genetically identical to the sheep that provided the donor egg and the donated nucleus.
- D. Dolly was not completely genetically identical to either the sheep that provided the donor egg or the sheep that provided the donated nucleus.

8 Consider the following processes that occur as part of the cell cycle and reproduction:

A *synthesis* → B *meiosis I* → C *meiosis II* → D *fertilisation* → E

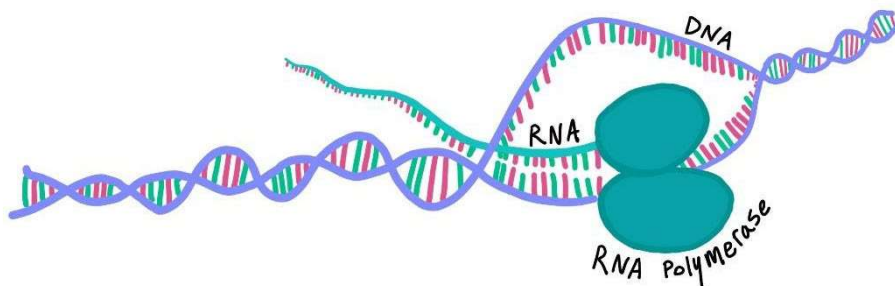
Which statement about the number of chromosomes is true?

- A. N chromosomes are present in an individual cell at stage A
- B. 4N chromosomes are present in an individual cell at stage B
- C. 2N chromosomes are present in an individual cell at stage C
- D. N chromosomes are present in an individual cell at stage D

9 When the temperature of the blood is above its set point, which of the following could occur?

- A. The pituitary gland releases a hormone to sweat glands to induce sweating.
- B. The hypothalamus sends a nervous signal to the muscles to rapidly contract.
- C. The pituitary gland releases a hormone to body cells to increase metabolic rate.
- D. The hypothalamus sends a nervous signal to arterioles to induce vasodilation.

10 Examine the following diagram:



Which of the following processes is being represented?

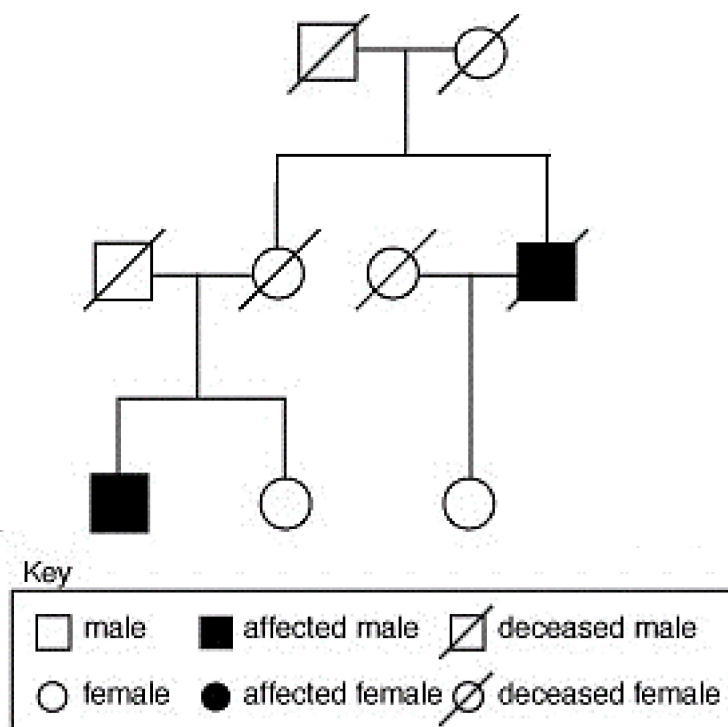
- A. Polymerisation
- B. Replication
- C. Transcription
- D. Translation

- 11** In humans, the rupture of an ovarian follicle occurs after a surge in which of the following hormones?
- A. Oestrogen
 - B. Progesterone
 - C. Luteinising hormone
 - D. Follicle stimulating hormone
- 12** After a careless laboratory injury involving laceration due to a broken test tube, a student observes later that afternoon that the cut is red, hot and swollen.
Which component of the immune system is most likely involved in this response?
- A. Plasma B cells
 - B. Cytotoxic T cells
 - C. Neutrophils
 - D. Antibodies

- 13 Mutation, gene flow and genetic drift can affect the gene pool of populations. Which of the following statements about the effects of these processes is true?

- A. Mutation and gene flow both increase the size of the gene pool.
- B. Mutation and genetic drift both increase the size of the gene pool.
- C. Gene flow and genetic drift both decrease the size of the gene pool.
- D. Mutation and genetic drift both decrease the size of the gene pool.

- 14 Consider the following pedigree chart for a hereditary condition:



The causes of death of the deceased individuals are not related to the heredity condition.

Individual III-3 is expecting a child with a male who is known to be a carrier of the condition.

What is the probability that the child will have the condition?

- A. 0%
- B. 25%
- C. 50%
- D. 100%

- 15** In order to investigate the microbes present in a range of water samples, a group of students carried out the following procedure:
1. 12 identical agar plates were made in sterile petri dishes.
 2. Samples of tap water, rain water and dam water were collected.
 3. 3 agar plates were sealed.
 4. An inoculation loop was heated in a Bunsen flame, cooled, inserted into the tap water sample then swiped over the surface of one of the remaining agar plates.
 5. The agar plate was then sealed.
 6. Steps 4-5 were repeated twice
 7. Steps 4-6 were repeated with the rain water sample and then the dam water sample.
 8. All plates were incubated for 3 days at 35 °C.
 9. Each plate was observed, and the number, appearance, size and shape of each colony was recorded.
 10. The agar plates were heated and then disposed of.

Which of the following statements about the procedure is true?

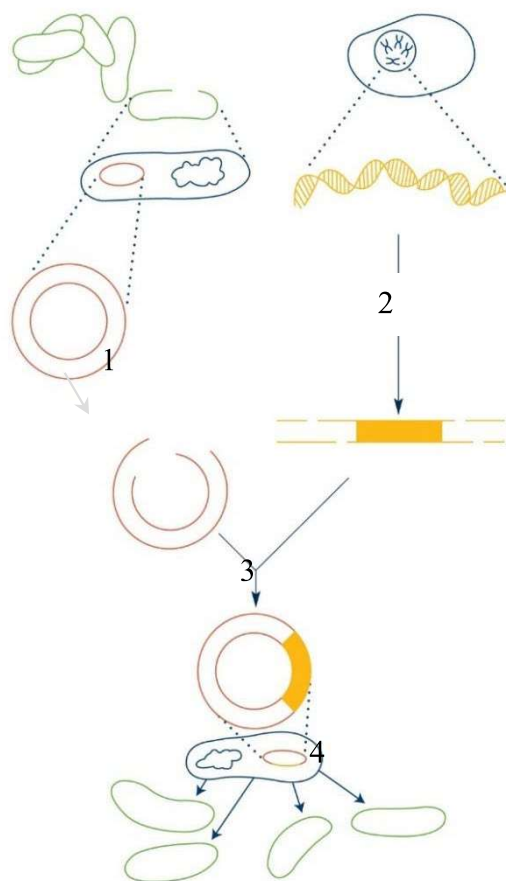
- A. Step 4 is important for ensuring reliability
 - B. Step 3 is important for ensuring controlled variables
 - C. Step 8 is important for ensuring a safe working environment
 - D. Step 1 is important for ensuring validity
- 16** The BRCA 1 gene is a tumour suppressor gene involved in the repair of double stranded breaks. Mutations to this gene are associated with significantly higher risks of cancer, in particular breast and ovarian cancer in women.

Mutations generally render the protein product non-functional. Although an individual only needs one functioning copy of this gene, the risk of developing these cancers is inherited in an autosomal dominant fashion. No patient has ever been observed to have inherited a mutated allele from both parents.

Which of the following statements is consistent with this information?

- A. A mutation in one allele means that the other allele has to work harder, increasing the risk of cancer.
- B. An interaction between the environment and the protein produced by the mutated BRCA1 gene determines whether or not the cancer develops.
- C. Damage to the functional allele is unable to be repaired properly, leading to a cancerous cell.
- D. Being homozygous for the mutation offers a level of protection against cancer.

Use the following diagram of DNA recombination to answer Questions 17 and 18.



17 Which step requires the addition of DNA ligase?

- A. 1
- B. 2
- C. 3
- D. 4

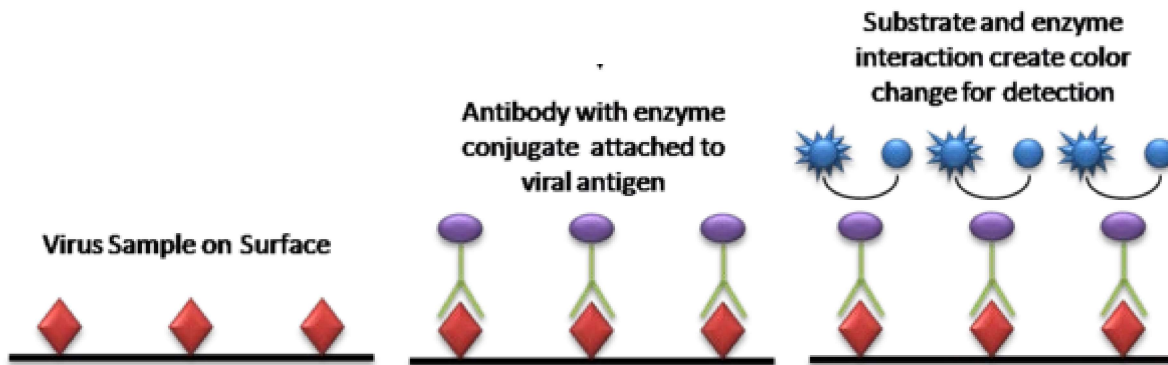
18 Antibiotic resistant genes are commonly included in the bacterial plasmid as part of the DNA recombination process.

Which of the following best explains the inclusion of antibiotic resistant genes?

	Fate of antibiotic resistant gene in successful DNA recombination	Purpose of antibiotic resistant gene
A.	Broken	To determine if recombination has occurred
B.	Broken	To determine if transformation has occurred
C.	Intact	To determine if recombination has occurred
D.	Intact	To determine if transformation has occurred

Use the following information to answer Questions 19-20.

Enzyme-linked immunosorbent assay (ELISA) is a technique that can be used to detect antigens. In the first step of an ELISA, antigens from the sample (e.g. a virus) to be tested are attached to a surface. Then, in the second step, a matching antibody is applied over the surface so it can bind to the antigen. This antibody is linked to an enzyme and then any unbound antibodies/enzymes are removed. In the third and final step, a substance containing the enzyme's substrate is added. If the antigen is present, the subsequent reaction produces a detectable signal, most commonly a colour change.



- 19 Which component of an ELISA is responsible for detecting a specific antigen?
- A. Antigen
 - B. Antibody
 - C. Enzyme
 - D. Substrate
- 20 Which component of an ELISA is responsible for the ability to detect an antigen at low concentration?
- A. Antigen
 - B. Antibody
 - C. Enzyme
 - D. Substrate

Section I**Multiple Choice Answer Sheet**

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|-----|----------------------------|----------------------------|----------------------------|----------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |
| 11. | A ☐ | B ☐ | C ☐ | D ☐ |
| 12. | A ☐ | B ☐ | C ☐ | D ☐ |
| 13. | A ☐ | B ☐ | C ☐ | D ☐ |
| 14. | A ☐ | B ☐ | C ☐ | D ☐ |
| 15. | A ☐ | B ☐ | C ☐ | D ☐ |
| 16. | A ☐ | B ☐ | C ☐ | D ☐ |
| 17. | A ☐ | B ☐ | C ☐ | D ☐ |
| 18. | A ☐ | B ☐ | C ☐ | D ☐ |
| 19. | A ☐ | B ☐ | C ☐ | D ☐ |
| 20. | A ☐ | B ☐ | C ☐ | D ☐ |

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Section II – 80 marks

Attempt Questions 21 – 31

Allow about 2 hour and 25 minutes for this part.

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Show all relevant working in questions involving calculations
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Question 21 (3 marks)

Describe ONE response of a named Australian plant to a named pathogen.

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Question 22 (3 marks)

Compare the features of bacteria and viruses.

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Question 23 (5 Marks)

Complete the following table for a named non-infectious disease.

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Name of disease	
Cause of disease	
Effect of disease	
1 st method of prevention	
2 nd method of prevention	

Question 24 (3 marks)

Outline the process of translation.

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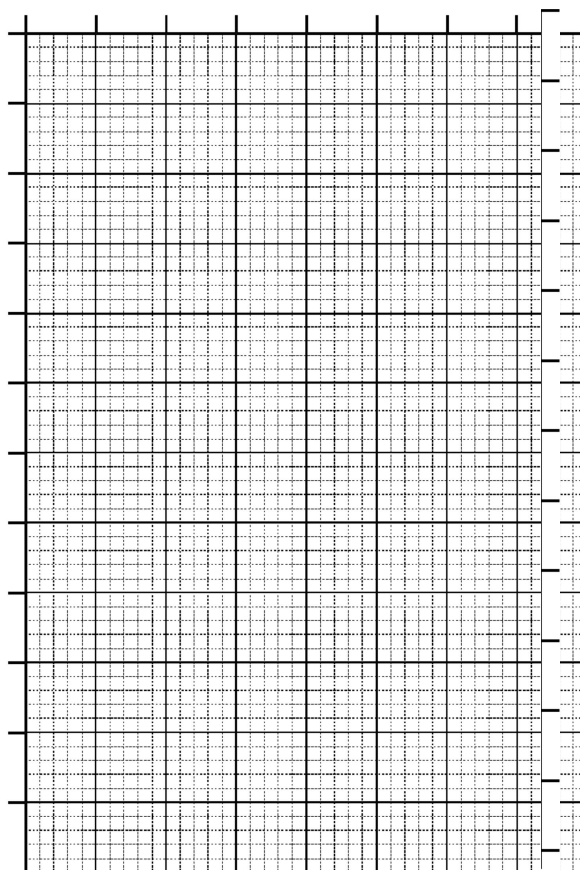
Question 25 (13 Marks)

The following data refer to the effect of soil temperature on the germination of onion seeds.

Temperature (°C)	% Seeds germinating	Time taken for seeds to emerge (days)
0	90	136
5	98	31
10	98	13
15	98	7
20	99	5
25	97	4
30	91	4
35	73	13
40	2	25

(a) Graph all the data on the grid provided.

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Question 25 continues on the next page

Question 25 (continued)

(b) Describe the trends in the graph.

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(c) With reference to your graph in part (a), discuss the optimal temperature for planting onion seeds.

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(d) Evaluate the impact of scientific knowledge on the manipulation of plant reproduction in agriculture using ONE example other than germination temperature.

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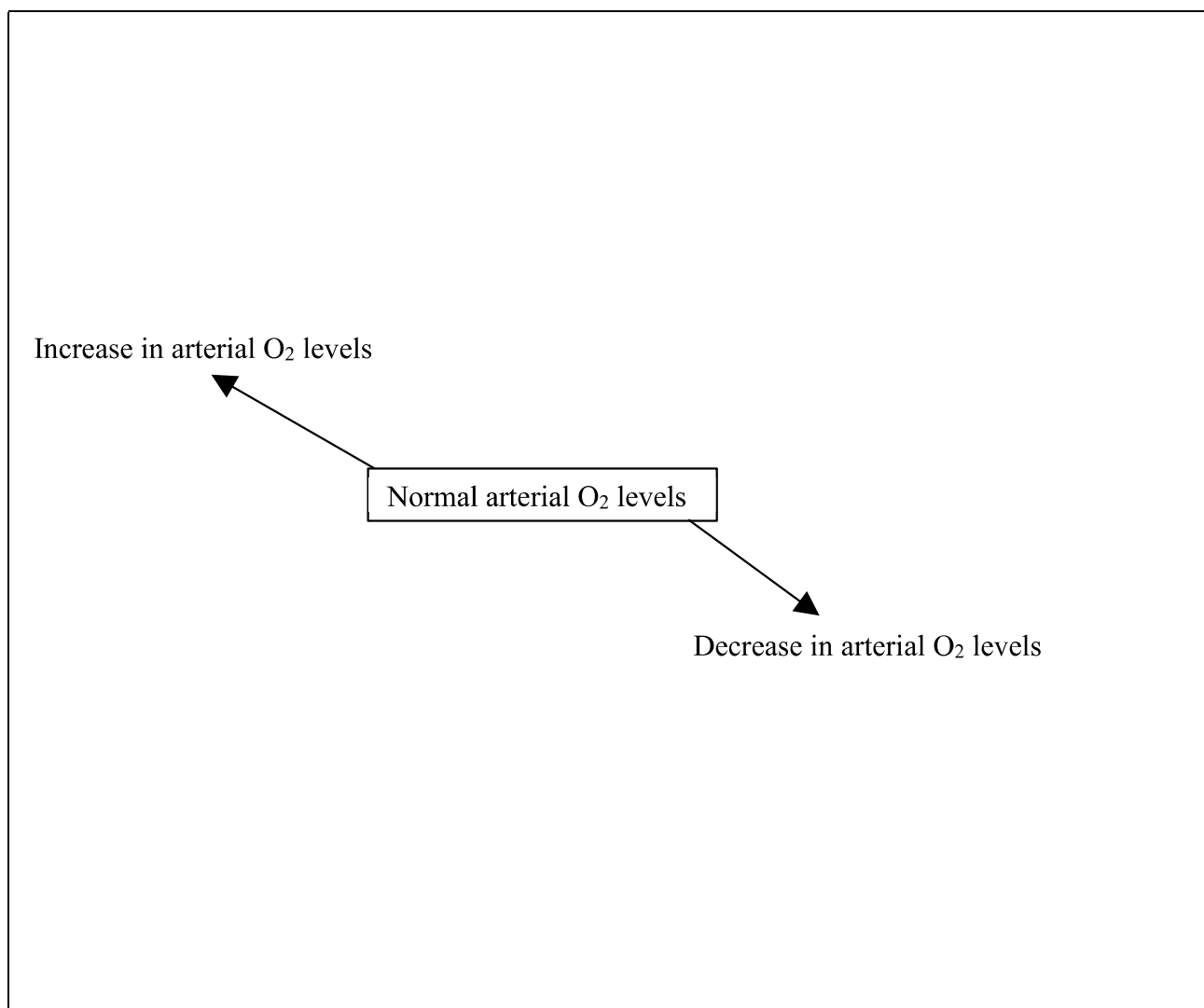
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Question 26 (8 Marks)

The average resting heart rate in an adult is about 70 beats per minute. The contraction of heart muscle cells is controlled by pacemaker cells which create rhythmic impulses. The pacemaker cells are controlled by the sympathetic nervous system, which acts to increase the heart rate and the parasympathetic nervous system, which acts to decrease the heart rate.

A number of factors affect the heart rate, in particular the partial pressure of oxygen in arterial blood. When low oxygen levels (hypoxia) are detected by chemoreceptors in the carotid body, a message is sent to cardiorespiratory centre in the medulla oblongata and the sympathetic nervous response is increased. When high oxygen levels (hyperoxia) are detected, the parasympathetic nervous response is increased.

- (a) Complete the following diagram showing the homeostasis of arterial oxygen levels, identifying stimuli, receptors, effectors and responses. **6**



Question 26 continues on the next page

Question 26 (continued)

- (b) Explain why the control of heart rate due to changes in oxygen levels is controlled by the nervous system rather than the endocrine system. **2**

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Question 27 (8 Marks)

The basic coat colours of horses include bay, chestnut and black.

The MC1R gene controls the production of red and black pigment. There are three alleles of this gene: E, e, and e^a. The e and e^a alleles are recessive to E. In individuals with the E allele, black pigment is produced. In those individuals without the E allele red pigment is produced instead.

The ASIP gene, also known as Agouti, controls the distribution of black pigment, if there is any black pigment. The dominant allele (A) restricts black pigment to the points of the horse (mane, tail, lower legs, ear rims), while the recessive form (a) distributes black pigment uniformly over the body.



- (a) Using the example above, explain the meaning of the term “multiple alleles”. 2

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- (b) Identify TWO possible genotypes of the bay horse, as shown above. 2

Possible genotype 1	
Possible genotype 2	

Question 27 continues on the next page

Question 27 (continued)

(c) A chestnut horse is being considered for a breeding program to produce black horses.

(i) Explain why genetic testing would be needed.

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(ii) Justify which of DNA sequencing or DNA profiling would be necessary to determine if the horse is suitable for the breeding program.

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Question 28 (10 Marks)

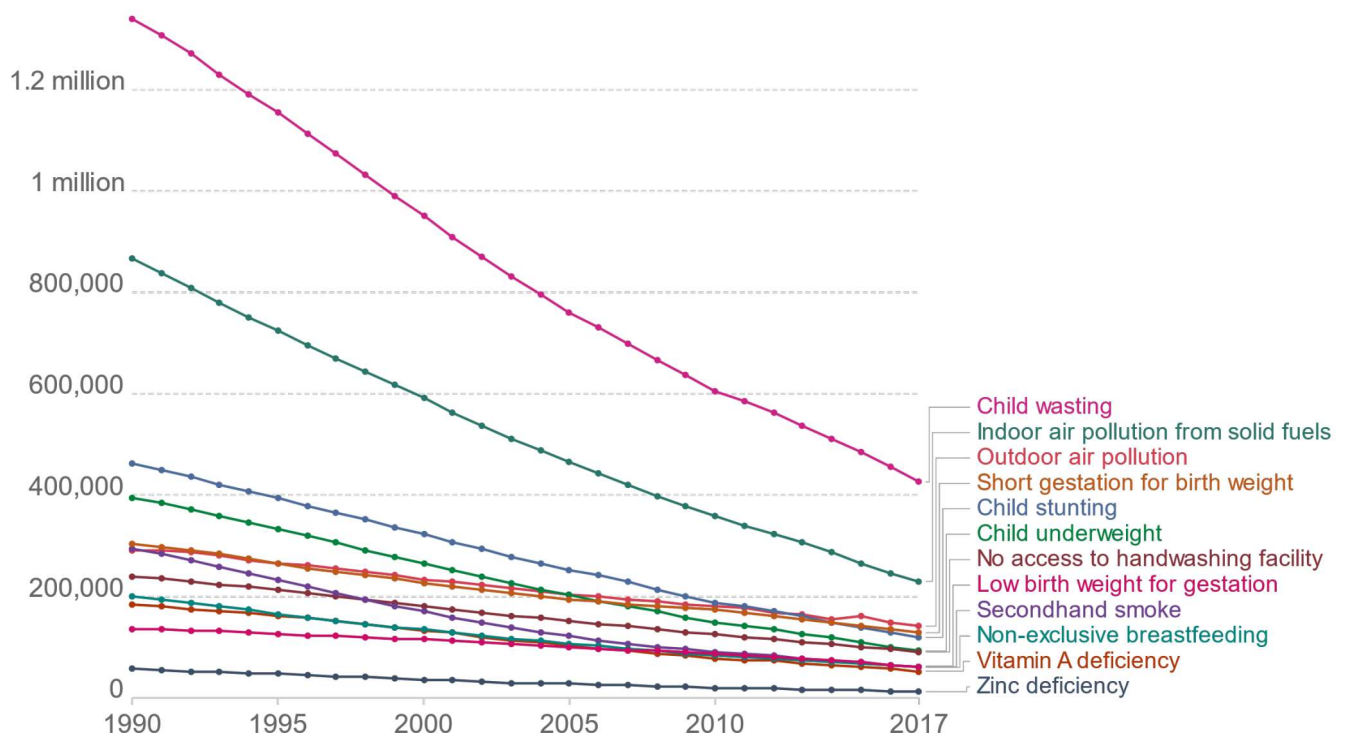
Pneumonia is an alveolar infection caused by a range of pathogens, including bacteria, viruses and fungi. The alveoli fill with pus and fluid, reducing oxygen intake and making breathing painful.

In 2017, pneumonia was the leading cause of death in children under the age of 5, responsible for over 800,000 fatalities, down from over 2 million per year in 1990. Although it is transmitted by air droplets, it is not highly contagious and is frequently the result of the consequences of poverty, with the largest risk factor having a weight too low for height (child wasting).

Child deaths from pneumonia by risk factor, World, 1990 to 2017

Our World
in Data

Shown is the number of deaths of children younger than five from pneumonia and other lower respiratory diseases by risk factor that was estimated to be responsible for these deaths.



Source: IHME, Global Burden of Disease Study (2017)

OurWorldInData.org/pneumonia • CC BY

Note: Risk factors can interact and overlap for a given cause of death; therefore the sum of deaths attributed to risk factors will exceed the total number of deaths from pneumonia.

Slightly more than half of fatal infections of children under the age of 5 were caused by the bacterium *Streptococcus pneumoniae*, which normally lives in the upper respiratory tract without causing illness. If the upper respiratory tract becomes compromised, then the bacterium can infect the lower respiratory tract, causing symptoms. A vaccine covering several strains of *Streptococcus pneumoniae* is available, which is 80% effective against the included strains.

Question 28 continues on the next page

Question 28 (continued)

- (a) Explain how the use of a vaccine may prevent an infection of *Streptococcus pneumoniae*.

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- (a) Using the data provided, justify THREE public health campaigns, apart from vaccination, that could significantly reduce the death rate of children under 5 due to pneumonia.

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Question 28 continues on the next page

Question 28 (continued)

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End of Question 28

Question 29 (3 marks)

Describe the use of ONE technology to assist with the effects of a non-infectious disease or disorder.

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Question 30 (7 Marks)

Epidemiological studies can be used to investigate both infectious and non-infectious diseases.

(a) Read the following extract from an article about the infectious disease diphtheria:

Diphtheria patients presented with fever, pharyngitis, and a patch in the throat. Data collected for each patient included age, clinical presentation, morbidity, mortality, bacteria isolated from culture, and immunization status.

Of the 61 950 admissions from January 2008 to December 2012, 2925 (4.7%) had clinical diphtheria; 1194 had been immunized and 1731 were non-immunized. Immunized patients had a milder disease. Culture-positive immunized patients were positive for *Corynebacterium* other than diphtheriae (COD; n = 104) or *Corynebacterium diphtheriae* (CD; n = 23); these patients suffered mild disease and recovered completely. In contrast, culture-positive non-immunized patients were positive for COD (n = 11) or CD (n = 412). Eighty-one patients (3%) died, 77 of whom were non-immunized; death was usually as a result of myocarditis. Seventy-three percent of deaths were in patients aged <5 years.

(a) Using the information above, complete the following table:

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	Infectious Disease
Study	Diphtheria in Andhra Pradesh
Type of study	
Summary of study findings	

Question 30 continues on the following page

Question 30 (continued)

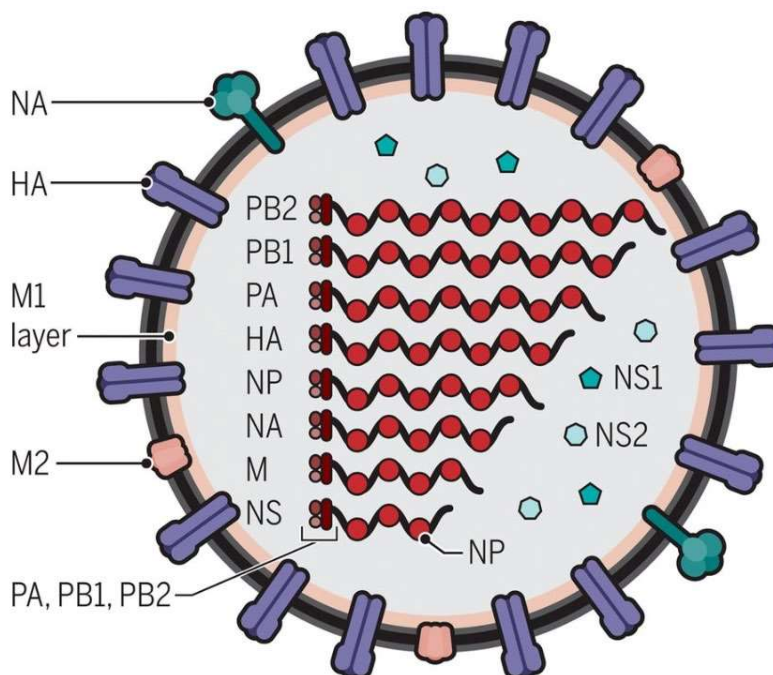
- (b) As part of your studies, you investigated an epidemiological study of a non-infectious disease.
Using your investigation complete the following table. **4**

	Non-infectious Disease
Study	
Type of study	
Summary of study findings	

Question 31 (17 Marks)

The influenza A virus is an RNA virus that infects birds and some mammals, including humans. There are a number of subtypes, described by the type of HA protein and NA protein e.g. H1N1, H5N2. Within a subtype there are multiple strains, which vary in their pathogenic ability between host species.

The genome of influenza A is spread across 8 segments and contains at least 10 genes as shown below in the diagram of the virus.



Each segment comprises single-stranded negative-sense RNA, which needs to be transcribed to a positive-sense mRNA using RNA polymerase before translation by the host. The genes PA, PB1 and PB2 together form the RNA polymerase. Each segment is transcribed in the host nucleus, with segment M containing two genes for matrix proteins (M1 and M2) and segment NS containing two genes for non-structural proteins (NS1 and NS2). The other segments each contain one gene.

The virus infects the host using surface HA proteins which bind to epithelial cells in the upper respiratory tract. These cells then import the virus by endocytosis. Once inside the cell, the viral RNA and proteins are released into the cytoplasm, then transported into the nucleus where viral replication and protein synthesis occur.

New virus particles form when the copied RNA and proteins enter a cluster of HA and NA which forms the new membrane and buds off from the host cell. The NA protein allows the newly-formed virus particle to leave the surface of the host cell and subsequently infect another cell.

Infected cells release a range of molecules which prevent viral entry into nearby cells and attract natural killer cells, macrophages and neutrophils. Dendritic cells present viral antigens to naïve T cells, and eventually antibodies are produced.

Question 31 continues on the next page

Question 31 (continued)

- (a) The RNA polymerase has no proof-reading function, leading to high rate of mutation, about one error per replication. This is termed antigenic drift. Immunity to one strain is likely to result in immunity to a mutated form when the difference in antigens is small.

Outline TWO possible consequences of point mutation in the coding regions of influenza A.

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- (b) In which of the influenza A genes is point mutation most likely to affect the ability of the adaptive immune system to combat the pathogen after a secondary infection?

Justify your response.

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Question 31 continues on the next page

Question 31 (continued)

- (c) If more than one type of influenza A virus particle infects a cell, different combinations of segments can combine to make new viral particles through reassortment. This is termed antigenic shift. For example, there are 18 variants of the HA gene and 11 variants of the NA gene. Immunity to one subtype is unlikely to result in immunity to another subtype. Recombination can also occur in an analogous manner to crossing over during meiosis.

New strains which become responsible for epidemics occasionally form in regions of poultry farming.

Propose how this might occur.

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Question 31 continues on the next page

Question 31 (continued)

(d) How important is it for genetic material to be replicated exactly?

Consider the lifecycles of both pathogen and host, using at least TWO examples from infectious diseases that you have studied.

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Question 31 continues on the next page

Question 31 (continued)

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End of Question 31

End of exam